Code No: 113AU **R13**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B.Tech II Year I Semester Examinations, March - 2022**
**ELECTRONIC DEVICES AND CIRCUITS**
**(Common to EEE, ECE, CSE, IT, MCT)**

**Time: 3 Hours** **Max. Marks: 75**

**Answer any five questions**
**All questions carry equal marks**
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1.a) Derive the expression for transition capacitance $C_T$ and Junction capacitance $C_J$ of a diode.
b) Justify the working of zener diode as a voltage regulator. [8+7]

2.a) Demonstrate the working of a fullwave rectifier with waveforms. How it is better than half wave rectifier?
b) A full wave rectifier circuit with C-type capacitor filter is to supply a D.C. Current of 25 mA at 16V. If frequency is 50 Hz and ripple allowed is 4%. Calculate:
i) Required secondary voltage of the transformer.
ii) Ratio of $I_{peak}/ I_{max}$ through diodes and the value of C required. [7+8]

3.a) Explain the input and output characteristics of a transistor in CB configuration. Mention one application of CB configuration.
b) Draw the circuit diagram of an NPN junction transistor in CE configuration and describe its characteristics. [7+8]

4.a) Discuss different stabilization methods against variations in $V_{BE}$ and $\beta$.
b) For the transistor amplifier circuit, when signal changes by 0.012 V, the base current changes by 9 µA and collector current by 1.3 mA. If the collector load $R_C$ = 5 KΩ, $R_L$ = 10 KΩ. Determine input resistance, current gain and voltage gain. [9+6]

5.a) Describe the operation of common drain FET amplifier and derive the equation for voltage gain.
b) Draw the VI characteristics curves of enhancement MOSFET and explain. Indicate different regions of operation on it. [7+8]

6.a) Justify how Varactor diode can be used as a variable capacitance?
b) Explain the working of Inductance filter connected at the output of a fullwave rectifier. Write all the corresponding waveforms. [7+8]

7.a) Illustrate and explain active region, saturation region and cut-off region in transistor characteristics.
b) Determine the quiescent currents and the collector to emitter voltage for a Silicon transistor with $\beta$ = 100 in the self biasing arrangements. The circuit component values are $V_{CC}$ = 12 V, $R_C$ = 2kΩ, $R_e$ = 0.1 kΩ, $R_1$ = 100 kΩ and $R_2$ = 5 kΩ. Find the stability factor S. [5+10]

8.a) Draw the JFET Small signal Model and explain it.
b) Compare BJT, JFET and MOSFET. [8+7]

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